On the Trade-off Between Benefit and Contribution for Clients in Federated Learning in Healthcare

1st Christoph Düsing
Semantic Computing Group
Bielefeld University
Bielefeld, Germany
cduesing@techfak.uni-bielefeld.de

2nd Philipp Cimiano
Semantic Computing Group
Bielefeld University
Bielefeld, Germany
cimiano@techfak.uni-bielefeld.de

***Abstract*—Federated Learning (FL) is a learning paradigm that allows clients to profit from the data that is available across multiple clients to train a joint model. As FL allows to train such a joint model without explicitly sharing data, but only sharing model updates, it has attained popularity in healthcare settings where patient data is subject to strict privacy policies and needs to be locally stored at each hospital or healthcare provider. A particular challenge for FL settings is data imbalance across clients, as it has been found to be detrimental to model performance and impact the influence of each client on the learning process. Unfortunately, the healthcare domain is particularly prone to such imbalanced data due to regional differences in disease management, prescription behavior etc. In this paper, we introduce the two novel metrics *Benefit* and *Contribution* to quantify to which degree individual clients benefit from participation in FL and how they contribute to its success, respectively. Therefore, we measure *Benefit* and *Contribution* with respect to four types of imbalances present in data at each client side. Our results show that both client *Benefit* and *Contribution* are influenced by data imbalance in such a way that high imbalance in data quantity, label distribution and feature distribution reduces or nullifies clients' *Benefit* while increasing their *Contribution*. Thus, the most valuable clients within a cohort benefit the least from their participation, exposing a critical thread to the success of clinical FL cohorts by withdrawing participation.**

***Index Terms*—Federated Learning, Healthcare, Imbalanced Data, Performance Evaluation**

I. Introduction

In recent years, applications of machine learning (ML) techniques in medicine have proliferated. ML based decision support systems are increasingly used to support diagnostic and therapeutic decisions. Typically, patient data available in so called electronic health records (EHR) is used to train such systems [1], [2]. It has been shown, however, that EHR data is fragmented [3] and shows a high regional variance [4] across providers. In fact, data imbalance might prevent individual providers to train ML models that generalize well [5], motivating them to join forces with other providers to train ML models with better performance. Federated learning (FL) is thus a promising paradigm that allows clients to profit from the benefits of a joint model while not having to share their data directly [6], [7]. In FL, clients can locally train a model on their own data sharing nothing but the model updates with a central party. This party aggregates all such model updates to a single FL model during several rounds of training [8]. However, it has also been shown that the performance of models trained in an FL setting suffer from data imbalance [9]. In fact, the influence of individual clients on the model has been shown to vary significantly [10]. Over time, numerous approaches have been proposed to improve performance and fairness of FL on imbalanced data. These approaches either improve the learning strategy on an algorithmic level [9], [11], [12], or seek to improve data quality and selection [13]. While results show that these approaches indeed improve the overall model performance, i.e., the performance on a test set representative for the entirety of data involved in the cohort, prior research fails to draw attention to individual clients and the performance achieved on their data.

Taking into account the benefit of individual clients that participate in the cohort is important, as in a healthcare setting the incentive for participation in a FL setting is typically to improve the own service quality [14], [15]. Participating providers are thus interested in using the joint model, a setting that we refer to as *usage-oriented* participation scheme [16]. In contrast, in the *compensation-oriented* participation, providers participate in the FL setting due to receiving financial compensation or in fulfillment of contractual obligations [17], [18].

We focus mainly on usage-oriented settings where participating clients expect a return-on-investment by receiving a better model in return of participating in the FL scheme [19], as it is the more common one in healthcare applications of FL. This motivates our work to go beyond the widespread evaluation of FL on a single test set that is representative for the entire data distributed among all clients. Instead, we draw attention to the margin of improvement each hospital can expect from participation.

According to Wang et al. [10], each client's participation is of different value for the cohort as well. Using a leave-one-out (LOO) analysis, they quantify the influence of each client on the FL model prediction, thus identifying the most valuable clients. While higher influence could increase compensation in a compensation-oriented setting, high-influence usage-oriented clients can only hope for large improvements in terms of performance. In this study, we therefore investigate potential correlations between benefit and contribution of individual clients to assess the fairness of compensation in usage-

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oriented, clinical applications of FL. More precisely, the underlying research question is: how does data imbalance affect clients' benefit and contribution in healthcare applications of FL and is there a trade-off between contributing to the success of FL and benefiting from participation?

Through empirical evaluations on different clinical datasets and data distributions among hospitals, we validate that the margin of imbalance among hospital data indicates the benefit of participation and find that high imbalance fosters less or no benefit. Moreover, we show that these high-imbalance hospitals are critical for the success of the FL, as their contribution is significantly higher than for low-imbalance clients. Ultimately, this necessitates to reevaluate clinical applications of FL and incentives for participation.

In summary, the main contributions of our paper are as follows:

1) We establish the need to draw attention to the benefit of individual clients in FL healthcare applications.
2) We propose a novel metric *Benefit* to quantify gain in performance for participating clients.
3) We propose the counterpart metric *Contribution* to quantify how much the shared model gains via the participation of a particular client.
4) We uncover a mismatch between *Benefit* and *Contribution* that potentially hinders successful future FL applications in the healthcare domain, as hospitals critical for the FL success might not benefit from participation in terms of performance.

In what follows, we present related work before introducing the methodology in Section III. Section IV contains results from our extensive experiments on two different datasets with different learning goals. Lastly, Section V concludes the paper and discusses results and future work.

II. Related Work

A. FL in Healthcare

Prior research identified FL as being particularly suited in healthcare applications where data privacy is a key criterion to opt for a distributed rather than centralized setting [3], which would require data to be collected at a single, central instance. Instead, FL allows hospitals and healthcare providers to participate by privately training a local ML model. During several rounds of local training and central aggregation of model updates, FL achieves a federated model that incorporates characteristics from data of all members of the cohort while keeping data private [20]. Applications of FL on EHR include mortality prediction [20], diabetes detection [21], understanding of clinical notes [22], tumor classification [7] and many others.

Although this comes at the cost of marginally decreasing model performance compared to central training [21], complying with the strict data privacy regulations [23] for EHR was found to outweight this drawback in typical healthcare applications.

B. Data Imbalance in FL

Although the quantity of available data is usually unproblematic, the imbalance of data among hospitals is. Past research on EHR of different healthcare institutions showed that regional differences in disease patterns and courses [4] as well as different fields of expertise [7] cause data within a FL cohort to be heterogeneous. In a study to improve cardiovascular diseases for example, Rao et al. [24] revealed that the number of patients with different diseases varies greatly. This prevalence of imbalance in real-world applications, e.g., with respect to target variables, can hinder model convergence and introduce biases [12].

This data imbalance poses a statistical challenge to potential FL cohorts [3]. To counter the negative impact of imbalance on FL, Li and colleagues [23] investigate the nature of imbalance. Therefore, they define three types of data imbalances, referred to as *skews*. These skews are *Label Imbalance*, *Quantity Imbalance*, and *Feature Imbalance*. Although neither skew exists in isolation in real-world applications (e.g. label imbalance is likely to correlate with feature imbalance), individually investigating these imbalances allows the authors to measure their respective effects on FL performance [23]. Unsurprisingly, presence of either type of imbalance causes the FL performance to decrease [23], [25]. To properly measure imbalance for individual clients and the entire cohort, Wang et al. [26] introduce the terms *global* and *local* imbalance. Global imbalance measures data imbalance for the entire FL cohort, whereas local imbalance refers to the data skew at each client side.

Other than that, different learning strategies are introduced improving the performance in face of imbalanced data. The original FL strategy *FedAvg* computes a weighted model average among all active clients in a round of FL [8]. However, FedAvg is particularly prone to suffering from data imbalance [25]. Therefore, different learning strategies have been proposed in order to better deal with its limitations. For example, Zhao and colleagues [9] propose to retrain the aggregated model each round on a shared, central dataset consisting of a small fraction of all clients' local data. More sophisticated approaches avoid this leakage of private data, e.g., by improving the local training process [27] or incorporating an attention mechanism to weight client contribution [28]. Overall, these learning strategies help to improve FL performance in specific domains but yet fail to solve the issue of data imbalance in general.

C. Client Benefit and Contribution

Up until today, only few studies have taken the client perspective into account when evaluating FL performance. However, evidence suggest that this is necessary, as a FL 'model may perform well on the global test but poorly on an individual clients's test data' [19]. Their study finds that the clients' benefit, i.e., the delta of accuracy for the FL model and a locally trained model, varies greatly and even drops below 0 for between 11% and 52% of clients within a cohort [19]. Rational, usage-oriented clients with little or no benefit might however be reluctant to participate in FL due to the loss of adequate incentives [19]. A complementary study further identified data imbalance as the main cause of this problem [29], but they did not measure the effect directly.

Except for decreasing FL performance, imbalanced data also affects the individual contribution of clients on the FL model [30]. This difference in contribution might be relevant to assess a fair compensation for FL participation and to identify malicious members within a cohort [31]. Thus, measurements of contribution, sometimes referred to as *Influence*, were established. Typically, this is achieved using LOO analysis [10], [31], where individual clients are removed from the cohort on each iteration. Afterwards, Wang et al. [10] measure the change in predicted probability distributions. In contrast, Huang et al. [31] have measured the delta of model accuracy. So far, contribution was used to identify malicious clients and to determine monetary compensation. However, it bears great potential for the analysis of fairness in usage-oriented FL applications. To the best of our knowledge, this has not been addressed in previous research.

III. Methods

Our methodology consists of four steps. First, we measure the local data imbalance of hospitals participating in FL. Thereafter, we quantify the benefit of each hospital by comparing the FL model to models trained at each local side. A similar idea is applied to measure the contribution of hospitals afterwards, measuring the change in FL model performance when certain hospitals are absent. Lastly, we provide a profound analysis of results and plot correlations of data imbalance, benefit and contribution.

Even though this study focuses on healthcare applications of FL, where clients are typically represented by hospitals or other healthcare providers, we stick to the common terminology of 'clients' instead of 'hospitals' for the remainder of this section, as the methodology presented is applicable to arbitrary applications of FL. However, from the experiments onward we will refer to participants of FL as 'hospitals' again.

A. Measure Data Imbalance

According to Li et al. [23], we consider three types of data imbalance: label imbalance, quantity imbalance and feature imbalance. Moreover, we stick to measuring local imbalance [26], as the goal of our work is to measure benefit and contribution for individual clients instead of success of the entire FL.

Label Imbalance. To quantify label imbalance, we define the two metrics *Label Imbalance* and *Label Distribution Imbalance*, inspired by Buda et al. [32] and Wang et al. [26]. These serve to address two distinct characteristics of imbalanced label distributions, i.e., the ratio of majority and minority class as well as the distribution of labels among clients. In accordance with Buda and colleagues [32], we define *Label Imbalance* LI_j of client j as

$$LI_j = \frac{max_p\{N_j^p\}}{min_p\{N_j^p\}}, \tag{1}$$

where p is the class index, and N_j^p the number of samples of the respective class at client j. In case of $min_p\{N_j^p\} = 0$, we define $LI_j = max_p\{N_j^p\}$.

Label Distribution Imbalance. To capture the mismatch in label distribution between clients and the cohort, we define $\vec{v}_j = [N_j^1, ..., N_j^Q]$ for each client j, where Q is the number of classes. Moreover, we define $\vec{V} = [\sum_j N_j^1, ..., \sum_j N_j^Q]$. Instead of using cosine similarity, as proposed in existing work [26], we define *Label Distribution Imbalance* as cosine distance (CD) of $\vec{v}_j$ and $\vec{V}$, i.e.,

$$CD_j = 1 - \frac{\vec{v}_j \cdot \vec{V}}{||\vec{v}_j|| \cdot ||\vec{V}||} \tag{2}$$

Thus, CD_j spans from zero ($\vec{v}_j$ is perfectly balanced with $\vec{V}$) to one (very high imbalance between $\vec{v}_j$ and $\vec{V}_j$), as all values in $\vec{v}_j$ and $\vec{V}$ are positive.

Quantity Imbalance. Inspired by previous measurements, we propose to extend exiting imbalance measurements and define *Quantity Imbalance* QI_j of client j as ratio of number of data points at client j and the mean number of data points over all clients. Formally, we define

$$QI_j = N_j / \frac{\sum_{l=1}^{J} N_l}{J}, \tag{3}$$

with J being the number of clients within the cohort. Accordingly, QI_j indicates to what extend the number of client samples diverges from the number of samples expected in a perfectly balanced setting.

Feature Imbalance. In order to account for *Feature Imbalance* FI_j -a nuance of data imbalance that has not been measured in previous research-, we propose to measure separability of the client data using a constrained k-means clustering on D with the number of clusters $k = J$. Here, D contains the data of all clients, defined as $D = \bigcup_{j=1}^{J} D_j$, with D_j being each client's local dataset. Using constrained k-means [33] allows to set the minimal cluster size to $s_{min} = 1$, avoiding empty clusters. The resulting cluster assignments C serve to compute the per-cluster purity. Thereby, we directly measure feature imbalance, as higher purity indicates high separability of the actual feature distribution among clients. Inspired by the *Purity* metric, we define FI_j as

$$FI_j = \frac{|C_j \cap D_j|}{|C_j|}, \tag{4}$$

where C_j is a set of instances assigned to cluster j.

Overall, we defined four metrics to measure data imbalance, namely *Label Imbalance*, *Label Distribution Imbalance*, *Quantity Imbalance*, and *Feature Imbalance*. For each metric it holds, that equalling zero indicates total absence of imbalance, i.e., homogeneity regarding the respective metric. Accordingly, higher values imply more severe imbalance.

B. Measuring Client Benefit

To take the clients' perspective into account it is necessary to quantify if and to what extent an individual client benefits from joining a FL cohort. To do so, we first train a FL model

θ^{FL} with the help of all clients. Moreover, a local model θ_j^{Local} is trained per client utilizing the same architecture as used in the FL process. Afterwards, we evaluate both models on the test data D_j^{Test} of the respective client that was held back from training entirely. We use *weighted F1-Score* (F1) to measure performance. Unlike previous research that relies on a single central test set, this allows us to assess FL in a usage-oriented clinical setting. Finally, inspired by Yu et al. [19] and Li et al. [29], we define *Benefit* as delta of F1 between FL and local model, i.e.:

$$Benefit_j = F1(\theta^{FL}, D_j^{Test}) - F1(\theta_j^{Local}, D_j^{Test}) \quad (5)$$

Consequently, for $Benefit_j > 0$, the respective client j observes an increase in the performance of the shared model on the own data when joining the cohort, otherwise the local learning objective might be too dissimilar from the global learning objective to profit from participation.

C. Measuring Client Contribution

Existing work on the success of FL models analyses the influence of individual clients [10], [31]. In particular, they perform a LOO analysis, iteratively removing individual clients from the cohort and retraining a FL model from scratch [10]. Both show the suitability of this approach for small cohorts. However, this approach entails two drawbacks when the cohorts are large: (1) it causes huge computational overhead, as the entire FL model has to be retrained for each client and, (2) the contribution of individual clients vanishes in large cohorts, making it indistinguishable from standard deviation. To avoid these limitations, we propose a leave-x-out (LXO) analysis instead. Here, we remove not one but several clients of similar kind from the cohort. Therefore, we assign each client a feature vector containing their previously measured imbalances $\vec{l_j} = [LI_j, CD_j, QI_j, FI_j]$ and apply another constrained k-means clustering on $\vec{L} = [\vec{l_1}, ..., \vec{l_J}]$. For $J < 10$, we set the number of clusters to $k = J$. For larger J, we set $k = J/5$. We constrain the clusters to have a size of $s_{min} = D/(2k)$ and $s_{max} = D/(0.5k)$. Finally, per cluster $c \in C$, we compute the mean of all four imbalances among the clients assigned to c and train the FL model θ_c^{LXO} without the participation of c.

To measure the actual effect of participation of clients, existing work defines *Influence* measuring the mean change in probability distributions for a dropped-out client [10]. However, this does not consider if the change in prediction caused by the respective client is actually beneficial for the cohort. Thus, Huang et al. [31] propose to measure the delta in *Accuracy* on a global testset D^{Test} instead. Thereby, they take into account the extent to which the cohort profits from a client's participation in terms of increased model performance. Our work refines existing influence measures by considering the F1 instead of accuracy, as it is widely considered the more meaningful and robust measurement. Hence, we define

$$Contribution_c = F1(\theta^{FL}, D^{Test}) - F1(\theta_c^{LXO}, D^{Test}) \quad (6)$$

The higher the resulting *Contribution* is, the more important is the respective cluster of clients for the success of the overall FL model.

IV. Experiments

A. Datasets, Problem Definition, and Model Architecture

In order to investigate the effects of data imbalance, we consider two publicly available healthcare datasets, namely *Diabetes* and *CTG*.

The Diabetes dataset contains features for over 100,000 diabetic patients [34]. To create a meaningful feature representation, we remove all variables with no clinical relevance such as case ids and admission information. The target class is binary, indicating whether a patient was provided a dose of insulin. For details on features and their descriptions, see [34]. The CTG dataset contains fetal cardiotocogram information and the fetal heart rate pattern class (1 to 10) assigned to each of them. For details on the features and their descriptions, see [35]. Further details about the datasets are provided in Table I.

TABLE I: Statistics of Datasets

Name	Instances	Features	Classes
Diabetes	101766	37	2
CTG	5791	22	10

For both learning tasks, we deploy simple neural networks consisting of a single hidden layer only. For both, the hidden layer is half the size of the input layer. The size of the output layer is defined by the number of classes for each dataset. The choice of model architecture is inspired by Li and colleagues [23], who also investigated FL performance on imbalanced, tabular data.

In order to apply these models in a federated setting, we deploy FedAvg [8]. Although more sophisticated learning strategies exist, FedAvg is the de facto standard FL strategy [23] used in many healthcare applications (e.g., [7]). Moreover, we split the datasets into a global testset and train data. Then, train data is split among J simulated hospitals. During splitting, we either assure homogeneity of data or enforce certain imbalances such as label, quantity or feature imbalance, as conducted in similar studies [23].

It is worth mentioning that our intention was not to tune either model architectures or FL strategy to achieve the highest possible F1 for the tasks presented because it is not the goal of this work to improve state-of-the-art performance of these models but to measure changes in performance instead.

B. Effects of Data Imbalance on Clinical FL

At first, we investigate the effects of imbalanced data on *Benefit* and *Contribution*. To do so, we apply FL twice on both datasets. Once, we randomly assign each datapoint to one of 50 hospitals, the other time we force the assignment to cause high label distribution imbalance. Then, we measure both *Benefit* and *Contribution* of all hospitals/clusters and present the results for *Diabetes* dataset in Figure 1.

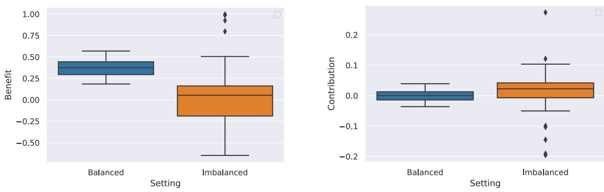


Fig. 1: *Benefit* and *Contribution* in Balanced and Imbalanced FL Settings

The results for the balanced FL setting reveal that (1) all hospitals have $Benefit_j > 0.2$ with little variance, meaning all of them profit from participation, and (2) the contribution closely varies around 0. In terms of imbalanced FL, the overall *Benefit* decreases when imbalance is high, whereas the variance of *Benefit* increases strongly. Most importantly a substantial fraction of hospitals show $Benefit_j < 0$. This implies that those hospitals do not benefit from participation at all. For *Contribution* in face of data imbalance, the effects are less severe. The plot shows that the median *Contribution* is marginally higher in case of high data imbalance and that there are more outliers with both positive and negative values.

All in all, the results show the potential of FL for the entire cohort when applied in a suitable environment. However, in an imbalanced setting, fewer hospitals get to profit from participation.

These results prove the necessity of further investigating the nature of these effects as conducted in the following sections.

C. Evaluation of Benefit

To shed light on the relationship between data imbalance and *Benefit*, we apply FL on both datasets while forcing certain data distributions among the hospitals. For the *Diabetes* dataset, we set $J = 50$ and for *CTG* $J = 25$, because of the significantly smaller number of data points. Then, we train θ^{FL} for 25 rounds each. The number of local epochs is set to 3. The local models θ_j^{Local} are trained for 50 epochs. After training is finished, we measure *Benefit* in accordance with Equation 5.

Figure 2 shows the results from our analysis; it plots the *Benefit* of each client using a linear regression line for each dataset and imbalance level. For *CTG*, we observe a clearly negative correlation between imbalance and *Benefit*, that is the higher the imbalance within a client the less the *Benefit* it has from participating in the FL scheme.

The results show strictly negative correlations of all imbalances and *Benefit*. In other words, those hospitals with higher imbalance exhibit lower *Benefit*. More precisely, *Quantity Imbalance* has the clearest negative impact on *Benefit*. This is unsurprising, as more data usually allows to train better models, whereas low-quantity data providers are incapable of training reasonable models themselves. However, the regression line of *Label Imbalance* bears high uncertainty and holds the potential of either strongly or marginally correlating with *Benefit*. For few hospitals, *Benefit* is even below zero, preventing usage-oriented participation, as locally trained models perform better on their own data. The findings regarding the *Diabetes* dataset are similar. Here Label Imbalance, Label Distribution Imbalance and Quantity Imbalance all show clear negative correlation with minor uncertainty. However, *Feature Imbalance* shows no clear correlation at all. Here, low Feature Imbalance comes along with both high and low *Benefit* values. Unlike for *CTG*, a much larger number of hospitals with highly imbalanced data was found to have $Benefit_j < 0$. This clear downwards trend of these hospitals with negative *Benefit* can be seen for *Label Distribution Imbalance*, where hospitals of highest imbalance also exhibit the lowest *Benefit*.

Overall, with only one exception, we find data imbalance to negatively correlate with *Benefit* of hospitals. Thus, it can be concluded that highly imbalanced hospitals are likely to benefit less - if at all - from FL participation.

D. Evaluation of Contribution

Next, we measure *Contribution* using a similar approach to previous analyses on *Benefit*. Again, we distribute data among 50 and 25 hospitals for *Diabetes* and *CTG*, respectively. Afterwards, we apply the LXO analysis of hospital *Contribution* by first measuring and then clustering clients' data imbalance vectors into $k = 10$ (Diabetes) and $k = 5$ (CTG) clusters. Afterwards, we train k FL models, iteratively leaving one cluster out. To increase reliability of our results, we repeat this procedure five times per dataset and plot the results in Figure 3.

Figure 3 shows that in fact, *Contribution* correlates with Data Imbalance. In the first row containing results for *Diabetes* dataset, we observe strong correlation of *Label Imbalance* and *Label Distribution Imbalance* with *Contribution*. This indicates that hospitals strongly deviating from the cohort's data distribution are more valuable than those showing less variation. The results for *Quantity Imbalance* and *Feature Imbalance* show mixed results. For Quantity Imbalance, we find a weak positive correlation with high uncertainty whereas for Feature Imbalance we measure weak negative correlation, again with some uncertainty. Regarding the *CTG* data, the figure shows that data imbalance of all kinds increases *Contribution*. In spite of the fact that some uncertainty remains, the positive correlation between imbalance and *Contribution* is clearly observable, that is, the higher the imbalance of a client, the higher its *Contribution* to the FL model performance.

All in all, the results allow to conclude that imbalance fosters *Contribution*, making hospitals with high imbalanced data most valuable for the cohort and critical for its success.

E. Correlation of Benefit and Contribution

Lastly, we aim to analyse the correlation of *Benefit* and *Contribution*. To do so, we assign each hospital the *Benefit* score we measured. Thereafter, we assign each hospital the *Contribution* score of their respective cluster divided by the number of hospitals in the cluster. Thus, we approximate the *Contribution* of each individual hospital to the FL process.

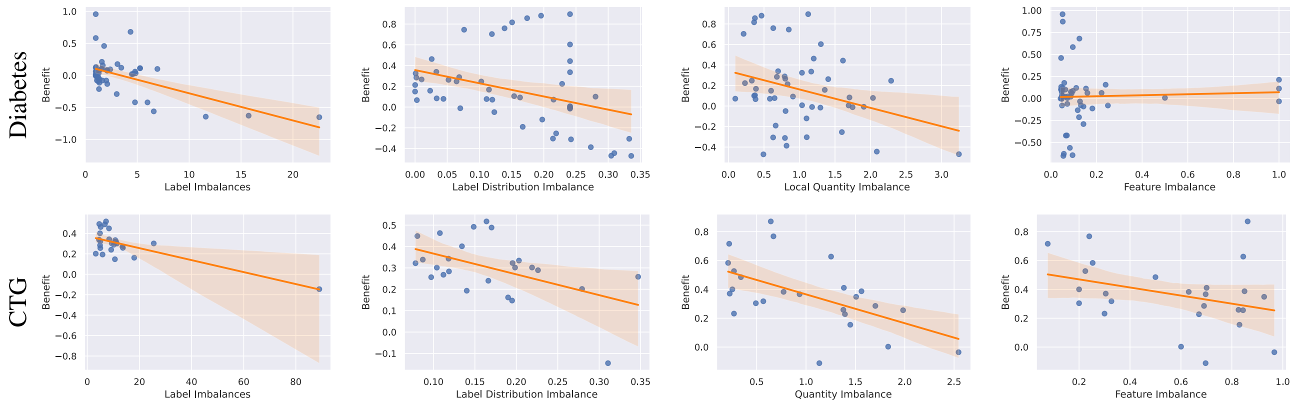


Fig. 2: Effects of Data Imbalance on *Benefit*

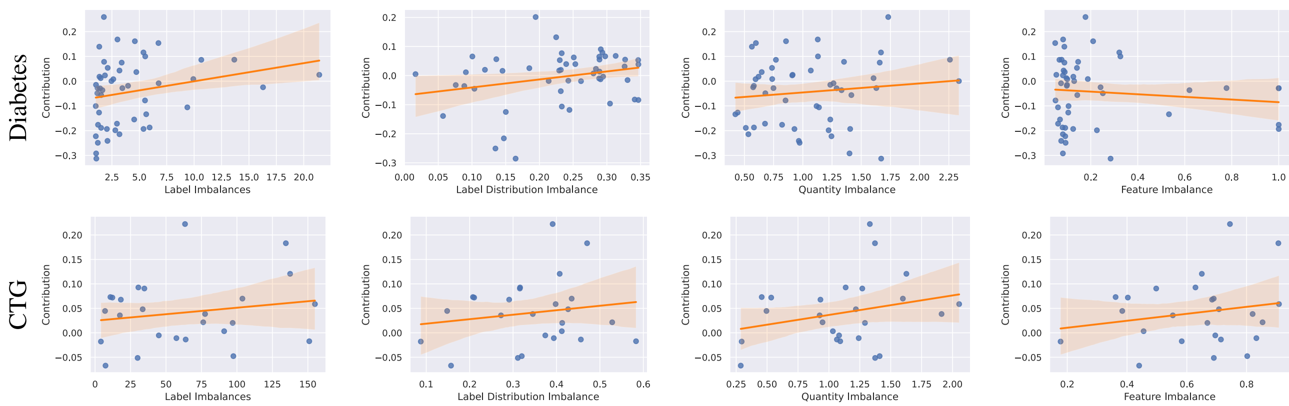
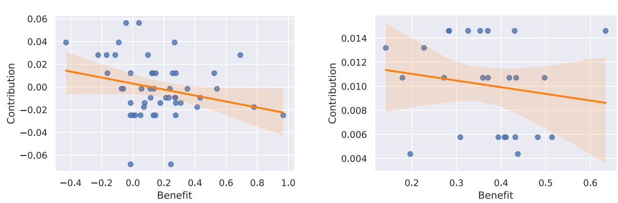


Fig. 3: Effects of Data Imbalance on *Contribution*



(a) Diabetes (b) CTG

Fig. 4: Correlation of *Benefit* and *Contribution*

This causes multiple hospitals to have the same *Contribution* score. Finally, we plot each hospital's *Benefit* and *Contribution* and complement the plots with a linear regression line.

From Figure 4 we observe a clear negative correlation between *Benefit* and *Contribution* on the *Diabetes* dataset. For *CTG*, we find a similar yet more vague correlation between the two metrics. In both cases, increasing *Benefit* correlates with decreasing *Contribution* and the hospitals in clusters of highest *Contribution* are far from being among those with the highest benefit. The results reveal a mismatch between being valuable for a FL cohort and getting to profit from participation, especially in usage-oriented, clinical applications of FL. A general conception of fairness would expect a positive correlation between the two variables.

V. Conclusion and Future Work

FL bears great potential for various applications within healthcare as it maintains data privacy and fosters collaboration and mutual support among healthcare providers. In our work, we show that FL -in a proper environment- entails great *Benefit* for the vast majority of hospitals, while hospitals provide similar *Contribution*. In an imbalanced setting however, hospitals with high imbalance benefit less from participation. In face of severe imbalances, locally trained models were found to even exceed the FL performance for individual hospitals. For such hospitals, usage-oriented participation is unfavorable. Regarding *Contribution*, we show the opposite. High imbalance fosters high *Contribution*. Thus, it is crucial for the cohort to keep these hospitals with highly imbalanced data although it would be reasonable for them to leave the cohort and rely on individually trained applications instead. Finally, we prove a trade-off between *Benefit* and *Contribution* for hospitals in FL. Unfortunately, this trade-off is favorable to low-imbalance clients only, whereas high-imbalance hospitals remain crucial for the success of the entire FL cohort.

In a wider perspective, our paper proves the necessity of future research to take both *Benefit* and *Contribution* of individual clients into account when evaluating FL performance. This is particularly relevant to the mostly usage-oriented healthcare domain, where hospitals seek to improve their own quality of service by participating in FL.

In the future, we aim to investigate the predictive potential of the imbalance measures for client *Benefit* and *Contribution*. Therefore, we have to (1) measure hospital data imbalance while maintaining privacy, and (2) generalize from task-specific measurement of both metrics. This would allow hospitals to better assess the likely *Benefit* from FL prior to actual participation.

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